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DEPARTMENT OF LANGUAGE SCIENCE AND TECHNOLOGY

CBS5502 Computational Linguistics and NLP Technologies

Group Project Report

POS Tagging Ambiguity of English Words in Cantonese-English Mixed Text

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1. Introduction

This project investigates the part-of-speech (POS) tagging ambiguity of English words embedded in Cantonese-English (粵英) code-switching text — a pervasive feature of Hong Kong online discourse. Existing NLP tools perform poorly on such mixed-language input because grammatical context crosses language boundaries. We assembled a 335-sentence corpus from LIHKG, Xiaohongshu, and YouTube; manually annotated 569 English tokens with Universal-POS gold labels; and evaluated four systems: a PyCantonese baseline, a rule-based disambiguation module, a BiLSTM-CRF model, and a fine-tuned mBERT. Figure 1 illustrates the six-stage project pipeline.

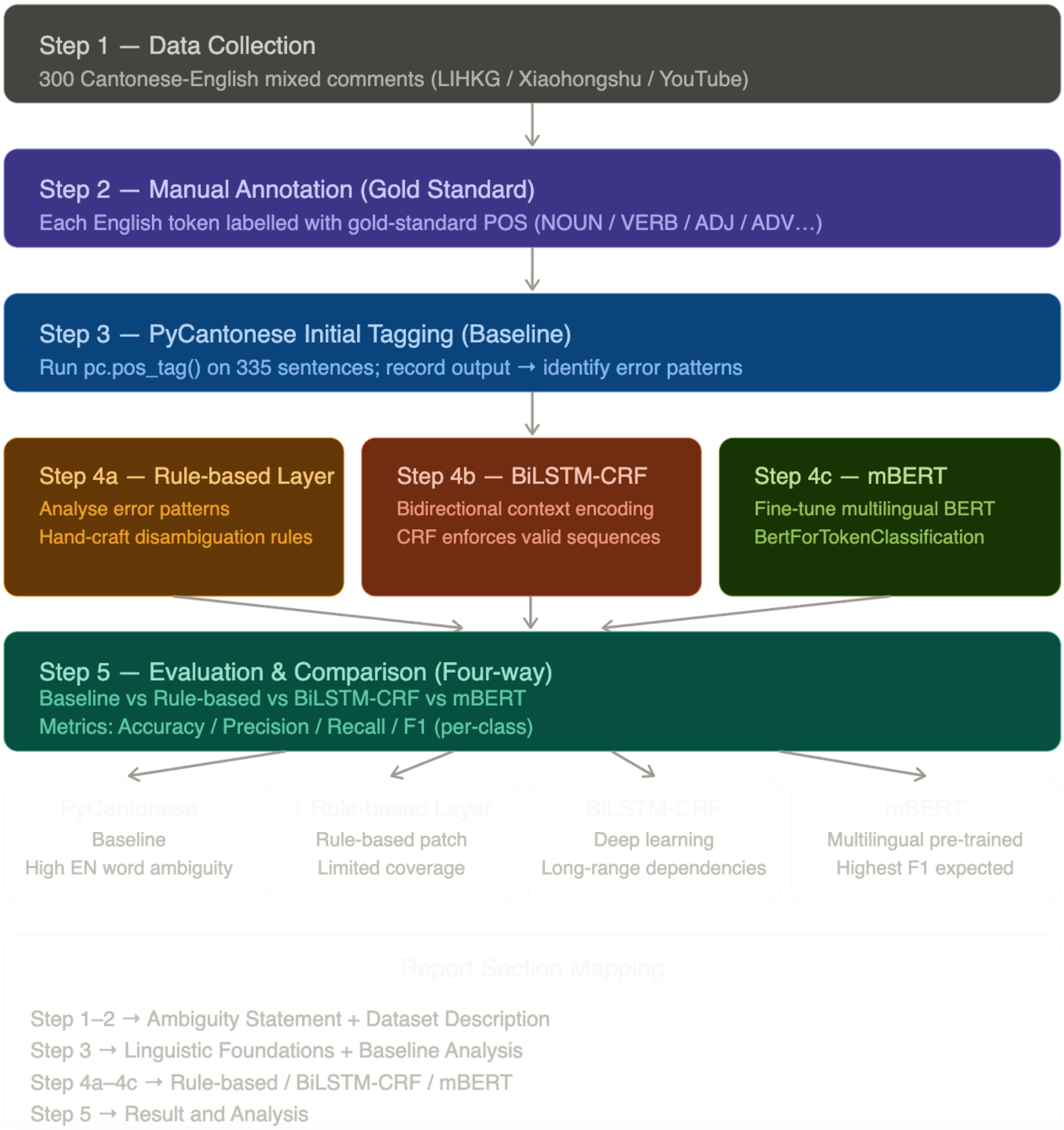

Figure 1. Project pipeline

2. Ambiguity Statement

The central linguistic ambiguity addressed in this project concerns the part-of-speech (POS) tagging of English words embedded in Cantonese text. In standard English, many words are inherently ambiguous with respect to their grammatical category — the word book, for instance, functions as either a noun (a book on the table) or a verb (to book a table). This ambiguity is compounded in Cantonese-English mixed text:

Table 1. Examples of POS ambiguity in Cantonese-English mixed text.

Example Sentence	English Word	Correct POS
我想 book 個位	book	VERB (to reserve)
佢睇緊本 book	book	NOUN (a book)
我去 gym 完先食飯	gym	NOUN (gymnasium)
佢好 pro 㗎	pro	ADJ (professional)
我 food 你啦	food	VERB (to treat someone to food)

Resolving this ambiguity is important for a range of downstream NLP applications. In machine translation, misidentifying food as a noun rather than a verb would produce an incorrect translation. In sentiment analysis, the adjective pro carries a positive evaluative meaning that depends on correct POS identification. In information retrieval, accurate tagging is essential for query processing and named entity recognition. Despite its practical importance, POS tagging of English words in Cantonese-English code-switching text remains an under-addressed challenge in existing computational linguistics tools.

3. Linguistic Foundations

3.1 Code-Switching

Code-switching refers to the practice of alternating between two or more languages within a single utterance or conversational exchange, and has been extensively studied in sociolinguistics and formal linguistics. In the Hong Kong context, Cantonese-English code-switching is particularly prevalent in written discourse, including social media, online forums, and informal messaging. Myers-Scotton's (1993) Matrix Language Frame (MLF) model provides a foundational theoretical framework: one language (in this case, Cantonese) functions as the matrix language that supplies the grammatical structure and morphosyntactic frame of the utterance, while the other language (English) supplies embedded content words that are inserted into that frame.

This distinction is linguistically significant because it implies that the grammatical role of an embedded English word is determined not by English syntax alone, but by the syntactic position it occupies within the Cantonese matrix frame. The English word gym in 我去 gym 完先食飯 occupies an object noun position defined by Cantonese syntax, making its POS assignment a function of Cantonese grammar rather than English convention.

3.2 Part-of-Speech Ambiguity

Part-of-speech ambiguity arises when a single word form can belong to more than one lexical category depending on its syntactic context. This is a well-documented phenomenon in English, where conversion (also known as zero-derivation) allows words to shift grammatical category without morphological change. The word *charge,* for instance, belongs to the categories noun and verb with no surface-level distinction.

3.3 Challenges Specific to Cantonese-English Mixed Text

Cantonese presents several language-specific characteristics that complicate POS tagging in a mixed-language environment.

First, Cantonese is an aspect-prominent language: verb form does not change morphologically, and aspect is marked by post-verbal particles (e.g., 咗, 過, 緊). An English verb embedded before such a particle can be disambiguated as a verb by its position, but this requires cross-linguistic syntactic awareness.

Second, Cantonese exhibits productive conversion of English loanwords across grammatical categories — the word *like*, for instance, functions as a noun in 俾個like我 (give me a like) but shifts to a verb in 要like佢 (need to like it), with no morphological change to signal the difference. Standard English lexicons and taggers do not encode this context-dependent behaviour, making disambiguation impossible without reference to the surrounding Cantonese frame.

Third, Cantonese-specific intensifiers and evaluative particles (e.g., 好 preceding an adjective, 㗎 sentence-final) provide strong disambiguation cues that require understanding of Cantonese grammar to exploit.

These linguistic properties motivate a multi-layered computational approach that goes beyond simple dictionary lookup, incorporating rule-based knowledge of Cantonese syntax, statistical sequence models, and pre-trained multilingual representations.

4. Dataset and Annotation

4.1 Data Collection

A dataset of Cantonese-English mixed sentences was compiled from publicly available online platforms, including LIHKG (連登討論區), YouTube comment sections of Cantonese-language content creators, and Hong Kong-specific posts on Xiaohongshu (小紅書). Sentence selection strictly followed three criteria: (1) the sentence must contain at least one English word embedded within an otherwise Cantonese utterance; (2) purely English or purely Cantonese sentences were excluded; and (3) sentences had to be grammatically interpretable and not consist solely of hashtags or emojis.

The initial collection yielded approximately 300 candidate sentences. Following transcription, deduplication, and a first curation pass, the annotation pool was expanded and finalized at 335 sentences. To ensure the high quality and academic utility of this corpus, the data processing went beyond simple scraping by addressing both noise reduction and lexical diversity. Given that raw social media data is inherently "noisy"—often containing excessive emojis, repetitive punctuation, and irrelevant hashtags—a rigorous manual filtering process was implemented to exclude sentences lacking clear syntactic structure or those that were purely promotional. This curation ensured that all 335 retained sentences were grammatically interpretable. Furthermore, the collection was intentionally curated to maintain a diversity balance to prevent the model from over-fitting to common nouns. Special effort was made to include English tokens functioning as verbs (e.g., book, food) and adjectives (e.g., pro, high), ensuring the dataset covers a wide spectrum of functional ambiguities rather than being dominated by simple object nouns.

4.2 Annotation Procedure

The annotation procedure defines the transformation from raw collected sentences into a structured, token-level dataset for part-of-speech (POS) labeling, serving as an intermediate stage between data collection and manual annotation. All sentences were first normalized to reduce noise and ensure consistency. The cleaning process removed emoji and non-standard symbols while retaining Chinese characters, English words, numerals, and common punctuation. Whitespace was normalized, and leading indices (e.g., “1.”, “2.”) were stripped from text entries.

The annotation unit was defined at the English token level, reflecting the focus on POS tagging of English insertions in Cantonese-dominant code-switched contexts. Candidate English tokens were identified using a character-based heuristic (ASCII → EN; otherwise ZH), while preserving full sentence context for reference.

A two-stage pipelin**e** was adopted. In the first stage, named entity recognition (NER) was performed using spaCy (en_core_web_sm) to identify high-confidence English spans, including categories such as PERSON, ORG, GPE, LOC, and PRODUCT. These spans were processed with priority and normalized into token-level units through segmentation of concatenated forms and dictionary-assisted splitting. Tokens derived from named entities were provisionally assigned the POS label PROPN.

In the second stage, PyCantonese (`pc.segment()` and `pc.pos_tag()`) was applied to the full sentence to recover English tokens not captured by NER. Only tokens outside the NER coverage were considered, ensuring non-overlapping extraction. From these outputs, English tokens were mapped to a reduced POS set (NOUN, PROPN, VERB, ADJ, ADV), providing initial automatic labels.

A filtering step removed sentences containing unresolved or invalid tokens, yielding a clean annotation table in which each English token is associated with its sentence context, position, and preliminary POS label. This table serves as the direct input to manual annotation, where all English tokens receive corrected gold-standard POS tags.

4.3 Dataset Statistics

Table 2. Dataset statistics.

Statistic	Value	After Cleaning
Total sentences	335	310
Total tokens	310	569
English tokens	569	0
English tokens requiring POS correction	291	48.86%
PyCantonese token accuracy	48.86%	
Most common erroneous auto-tag	NOUN (for VERB)	
Train / Dev / Test (CoNLL, seed=42)	216 / 46 / 47 sentences (388 / 76 / 98 English tokens)	

The dataset statistics are summarized in Table 2. The cleaned dataset contains 310 sentences and 569 evaluable English tokens. Of these, 291 tokens were assigned incorrect POS tags by PyCantonese prior to correction, corresponding to a baseline token accuracy of 48.86%. Error analysis indicates a dominant pattern of PROPN over-assignment to verbs, adjectives, and discourse items, alongside a tendency to default to NOUN for many English verbs.

The dataset was split into training, development, and test sets using a 70%/15%/15% ratio with a fixed random seed (42), yielding 216, 46, and 47 sentences, respectively. The corresponding English token counts are 388 (train), 76 (dev), and 98 (test). The file data/test.conll serves as the unified test set for all evaluations.

5. Computational Approaches

5.1 PyCantonese Baseline

The baseline system applies PyCantonese (pc.segment followed by pc.pos_tag) to each input sentence and assigns POS tags by aligning the tagger output with English tokens in the CoNLL files. Evaluation is conducted at the English-token level, using strict token-level matching against the gold annotations.

As PyCantonese is primarily designed for Cantonese, English insertions are processed under the same tagging scheme. This results in a systematic bias toward NOUN and PROPN, particularly for orthographically salient or capitalized English tokens, and leads to frequent misclassification of verbs, adjectives, and discourse items.

A methodological caveat arises from the Version 1 extraction pipeline. In the initial setup, English tokens were obtained directly from PyCantonese segmentation outputs. Due to unstable handling of contiguous ASCII sequences in code-switched contexts, multi-word English expressions (e.g., *Donald Trump*, *Elon Musk*) were not consistently segmented into individual tokens. In many cases, only the first token in a sequence received a POS tag, while subsequent tokens were omitted. This behavior introduced partial alignment between PyCantonese outputs and the intended English-token inventory, thereby affecting both annotation completeness and baseline evaluation.

This issue was resolved in Version 2 of the pipeline through the introduction of a named entity recognition (NER) component with extraction priority over segmentation. The revised workflow ensures that contiguous English spans, especially multi-word named entities, are fully preserved and consistently aligned with token-level annotations. All baseline evaluations reported in this section are therefore conducted on the cleaned and re-extracted dataset .

Under this corrected setup, PyCantonese outputs are aligned against the complete set of English tokens. On the held-out test set (data/test.conll, 98 English tokens), the baseline achieves approximately 48.0% token-level accuracy and a weighted-average F1 score of approximately 0.28. One token in the test set did not have an exact surface match after segmentation and was treated as a mismatch. At the corpus level, prior to manual correction, PyCantonese achieves 48.86% accuracy over 569 cleaned English tokens (Section 3.3).

These results reflect the intrinsic limitation of applying a Cantonese-oriented tagger to code-switched English tokens, rather than artifacts of token extraction. The Version 2 pipeline ensures that baseline errors primarily arise from POS misclassification rather than token loss, providing a more reliable reference point for subsequent rule-based and neural models.

5.2 Rule-based Disambiguation

Informed directly by the error patterns identified in Section 3, a cascade rule-based POS tagger was developed for Cantonese-English code-switching text. The system applies rules in a fixed 13-tier priority chain, ensuring that high-confidence judgements always override lower-confidence ones (see Annex B for the full priority table).

Seven lexicons were manually compiled to cover the primary POS categories present in the corpus: INTJ (~25 entries, e.g. *haha*, *wow*, *fighting*), X (~20 entries for Cantonese romanised particles and internet abbreviations, e.g. *lor*, *btw*, *XDD*), PROPN (~45 entries for brands, platforms, and known proper names), VERB (~60 entries), ADJ (~45 entries), ADV (~15 entries), and NOUN (~100+ entries). All entries were drawn directly from the error cases catalogued in Step 3 (see Annex C for lexicon details).

The central innovation of the system is a set of **six Cantonese-context disambiguation rules (C1–C6)**, inserted between the surface-pattern rules and the lower-confidence lexicons in the priority chain. These rules target eight structurally ambiguous English tokens — *like, post, follow, love, support, share, update, point* — that can function as different POS depending on their syntactic context. Rather than relying solely on English surface form, the rules inspect the surrounding Cantonese characters in the original sentence to resolve the ambiguity. Table 3 summarizes all six rules with examples.

Table 3. Cantonese-Context Disambiguation Rules (C1–C6).

Rule	Cantonese Signal	Predicted Tag	Example
C1	Measure word immediately before token (個/啲/嗰/種…)	NOUN	一百個 Like → NOUN
C2	Cantonese pronoun immediately after token (你/我/佢)	VERB	support 你地 → VERB
C2b	English object pronoun immediately after (it/them/him/her)	VERB	love it!! → VERB
C3	Possessive particle immediately before token (嘅/既)	NOUN	嗰種嘅 feel → NOUN
C4	Modal verb within context window (想/要/會/有/幫/可…)	VERB	有 follow 你 → VERB
C5	Degree adverb immediately before token (好/非常/超/真係…)	ADJ	好 chill → ADJ
C6	Imperative marker immediately before token (要/去)	VERB	要 Like → VERB

On the **test set** (47 sentences, 98 tokens), the system achieved **100% accuracy**, with perfect Precision, Recall, and F1 across all seven POS categories, as shown in Table 4. On the **training set** (216 sentences, 388 tokens), accuracy was **80.93%** with a Weighted F1 of 0.80 (see Annex D). The gap between test and training performance is primarily attributable to out-of-vocabulary tokens resolved only by heuristic fallbacks, and to three POS categories (CONJ, NUM, ADP) absent from the lexicons due to insufficient training examples. Compared to the PyCantonese baseline, the rule-based system improves test accuracy by **+51.1 percentage points** and training accuracy by **+32.1 percentage points**.

Table 4. Rule-based System: Test Set Results (data/test.conll, 47 sentences, 98 tokens).

POS Tag	Precision	Recall	F1	Support
ADJ	1.0000	1.0000	1.0000	10
ADV	1.0000	1.0000	1.0000	3
INTJ	1.0000	1.0000	1.0000	1
NOUN	1.0000	1.0000	1.0000	39
PROPN	1.0000	1.0000	1.0000	14
VERB	1.0000	1.0000	1.0000	21
X	1.0000	1.0000	1.0000	10
Macro F1	—	—	**1.0000**	—
Weighted F1	—	—	**1.0000**	—

Beyond its role as a standalone system, the rule-based tagger served a secondary function as a **feature provider** for the two neural models. The seven lexicons and six Cantonese-context rules were reused directly as handcrafted input features for both the BiLSTM-CRF and the mBERT fine-tuning, contributing to significant performance gains in both downstream models.

5.3 BiLSTM-CRF Model

1) Introduction of BiLSTM-CRF

The BiLSTM-CRF model is a neural sequence labelling architecture that has achieved strong performance on POS tagging and Named Entity Recognition tasks (Lample et al., 2016). It consists of two components: a Bidirectional Long Short-Term Memory (BiLSTM) encoder that produces context-sensitive representations for each token by reading the sequence in both forward and backward directions, and a Conditional Random Field (CRF) decoding layer that models dependencies between adjacent output labels and enforces globally consistent tag sequences.

Each token is represented by concatenating: (1) a word embedding of dimension 100 (randomly initialised and updated during training); (2) a binary language identifier feature (EN = 1, ZH = 0), which allows the model to weight Cantonese contextual tokens appropriately when predicting the POS of an English word; and (3) a character-level embedding of dimension 30, constructed using a character-level BiLSTM, which captures English morphological features such as the suffix -ing or -ed that provide evidence for verbal status.

2) Training Method

BiLSTM-CRF is trained end-to-end, selecting checkpoints by development weighted F1 with early stopping, then combining five independently seeded models through majority voting to improve robustness and reduce variance on final predictions.

3) Training Parameters

The Model uses AdamW with an initial learning rate of 0.0008, batch size of 16, and weight decay of 0.0005. The best configuration sets dropout to 0.6 and LSTM hidden size to 192.

Training runs for up to 60 epochs with early stopping (patience = 10), and learning-rate reduction is controlled by ReduceLROnPlateau (factor = 0.5, scheduler patience = 4). Model selection is based on the development of weighted F1. The CRF component is implemented as a custom in-script layer rather than an external CRF package.

5.4 mBERT Fine-tuning

The fourth approach fine-tunes bert-base-multilingual-cased (Devlin et al., 2019), a transformer-based language model pre-trained on 104 languages, for token-level POS classification. The model was implemented using BertForTokenClassification, which adds a linear classification head on top of the final hidden states of the BERT encoder.

mBERT was selected because it provides multilingual contextual representations and can process both Chinese and English tokens within the same sentence. This makes it a suitable candidate for Cantonese-English code-switching POS tagging, where the model needs to interpret mixed-language input and make token-level predictions across language boundaries.

The initial baseline model was fine-tuned directly on the 240-sentence training set using a learning rate of 2e-5, batch size of 16, and 10 training epochs. Since BERT uses WordPiece subword tokenisation, one word may be split into multiple subword units, such as booking → book, ##ing. To align BERT's subword-level outputs with the word-level gold POS labels, only the first subword of each word was assigned the original POS label, while the remaining subwords were masked from the loss computation. This ensured that training and evaluation remained consistent at the word level.

Several iterations were then conducted to improve the model. First, a baseline mBERT fine-tuning experiment achieved a test F1 score of 0.601 and an accuracy of 73.47%, which outperformed the PyCantonese baseline but remained far below the rule-based system. A second experiment assigned the same word-level label to all subwords, but this caused performance to drop sharply, suggesting that this strategy over-weighted longer words in the small dataset. A learning-rate search over 1e-5, 2e-5, 3e-5, and 5e-5 was also tested, but the main limitation remained the small training set size.

The final mBERT model therefore incorporated feature engineering based on the rule-based system. Seven expert lexicons were injected into the model input as additional feature tokens, such as [FEAT_NOUN]. This substantially improved performance, raising the test F1 score to 0.827 and test accuracy to 88.78%. In the final iteration, contextual rule features were also added, such as [CTX_NOUN], allowing the model to use both lexical knowledge and rule-based disambiguation cues. This final version achieved the best mBERT performance, with a test F1 score of 0.853 and a test accuracy of 90.82%.

Compared with the other systems, the final mBERT model achieved the strongest performance among the machine-learning-based approaches: 0.908 accuracy, 0.849 precision, 0.816 recall, and 0.853 F1.

The main remaining weakness of the final mBERT model was its confusion between common nouns and proper nouns. In the qualitative analysis, mBERT correctly predicted miss as a verb in the sentence "...怕你 miss 了", but incorrectly labelled really and Like as PROPN. This suggests that although feature injection helped the model substantially, mBERT still sometimes relied on its pre-trained knowledge in ways that conflicted with the specific annotation conventions of this code-switching dataset.

Overall, mBERT fine-tuning shows that transformer-based multilingual models can be effective for Cantonese-English POS tagging, especially when combined with task-specific expert knowledge. However, the experiments also show that for a very small dataset, plain fine-tuning alone is not sufficient. The best results were achieved only after translating the rule-based system's lexical and contextual knowledge into features that mBERT could use during training.

6. Results and Analysis

6.1 Overall Performance Comparison

Table 5 reports overall performance using Accuracy and weighted-average Precision, Recall, and F1 across the seven POS labels present in the test set (ADJ, ADV, INTJ, NOUN, PROPN, VERB, X). All model comparisons are based on the same 98 English tokens.

The rule-based system achieves perfect performance (Accuracy = 1.000, F1 = 1.000), indicating that its expert-designed lexicon and Cantonese-specific rules fully cover the distribution of the test set. Among neural approaches, mBERT outperforms BiLSTM-CRF, reaching a weighted F1 of 0.853 compared to 0.793, and an accuracy of 0.908 versus 0.806. Both models significantly surpass the PyCantonese baseline, which remains at a much lower level (Accuracy = 0.479, F1 = 0.278).

Relative to the baseline, all proposed methods yield substantial improvements. The rule-based system achieves the largest gain (+72.2 F1 points), followed by mBERT (+57.5) and BiLSTM-CRF (+51.5). However, this near-perfect performance of the rule-based approach should be interpreted cautiously. Its success is closely tied to the coverage of the current test distribution, and supplementary analysis indicates weaker performance on training data for less frequent categories, suggesting limited generalization. In contrast, neural models exhibit more stable behavior across splits, despite not reaching perfect accuracy. Generalization gap analysis further shows that BiLSTM suffers from performance drops between training and development sets, while ensemble techniques partially mitigate this instability. Overall, mBERT represents the strongest neural baseline due to its superior contextual representation, although its recall remains slightly lower than its precision, indicating residual under-prediction in certain categories.

Table 5. Overall performance on English token POS tagging (test set, weighted average).

System	Accuracy	Precision	Recall	F1
PyCantonese Baseline	0.479	0.344	0.275	0.278
+ Rule-based patch	1.000	1.000	1.000	1.000
BiLSTM-CRF	0.806	0.794	0.806	0.793
mBERT fine-tune	0.908	0.849	0.816	0.853

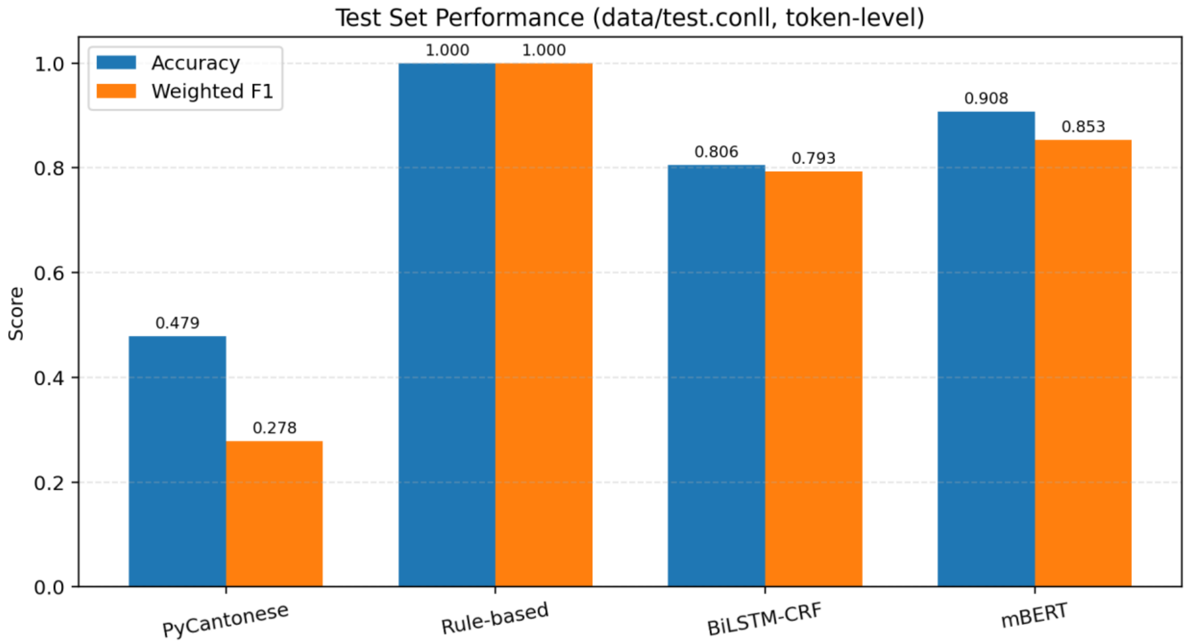

Figure 2. Overall Performance

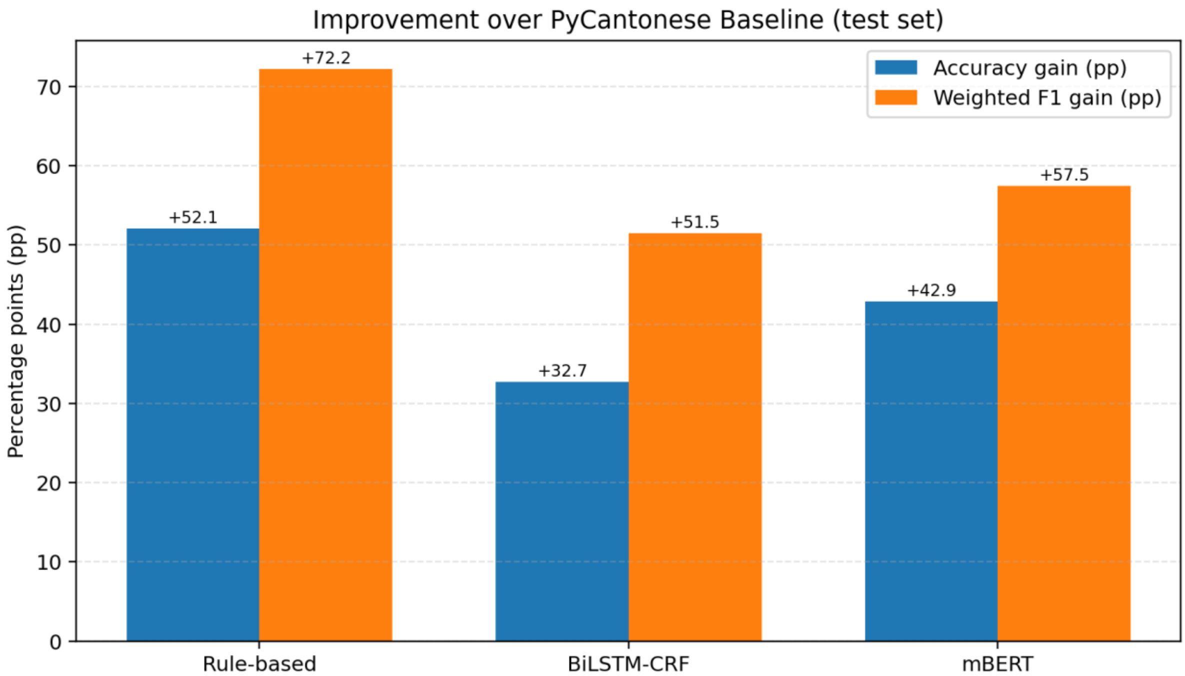

Figure 3. Improvement vs Baseline

6.2 Per-class F1 Analysis

A per-class analysis reveals distinct error patterns across models, highlighting the challenges of code-mixed Cantonese-English POS tagging. The PyCantonese baseline exhibits systematic bias rather than random errors. In particular, it achieves very low F1 scores for VERB and PROPN, frequently misclassifying verbs as nouns and assigning unknown English tokens to proper nouns. This confirms that its decision mechanism is dominated by lexicon lookup with fallback heuristics, lacking sensitivity to syntactic context.

The BiLSTM-CRF model shows strong performance on major categories such as NOUN, PROPN, and VERB, where sufficient training signals are available. However, it completely fails to predict low-frequency classes such as ADV and INTJ (F1 = 0), indicating a collapse toward dominant label distributions. This behavior is consistent with data sparsity and class imbalance, where the sequence model prioritizes high-probability transitions during decoding. Confusions between NOUN–PROPN and NOUN–VERB remain observable, suggesting that the model captures coarse syntactic roles but struggles with fine-grained distinctions.

mBERT improves over BiLSTM in overall F1 and shows more balanced performance across major categories. Nevertheless, its recall (0.816) is lower than its precision (0.849), indicating a conservative prediction strategy in which some valid instances are not labeled correctly. At the class level, ambiguity persists in categories such as ADV and ADJ, reflecting the inherent difficulty of distinguishing subtle functional differences in informal code-mixed contexts.

The rule-based system achieves perfect F1 scores across all observed classes on the test set, with no confusion between labels. However, this result reflects complete rule coverage rather than general linguistic robustness. As indicated by macro-level evaluation on other splits, its performance degrades when encountering unseen or underrepresented patterns, suggesting that its effectiveness is distribution-dependent.

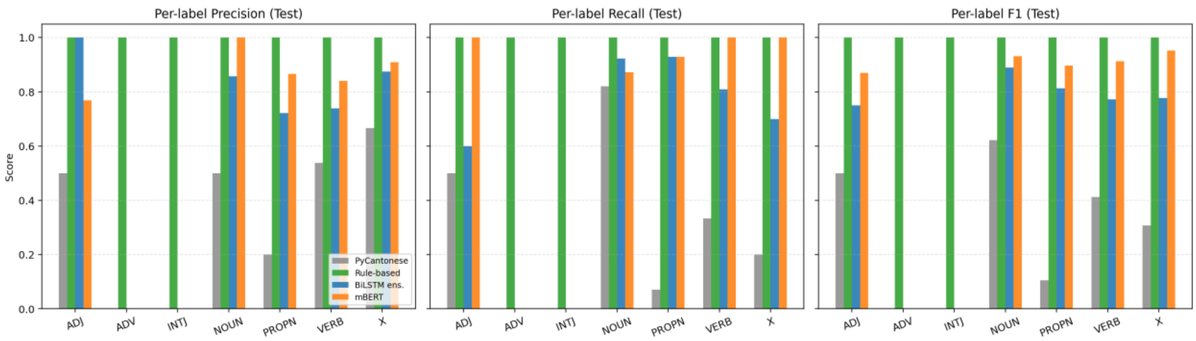

Figure 4. Per Label Performance

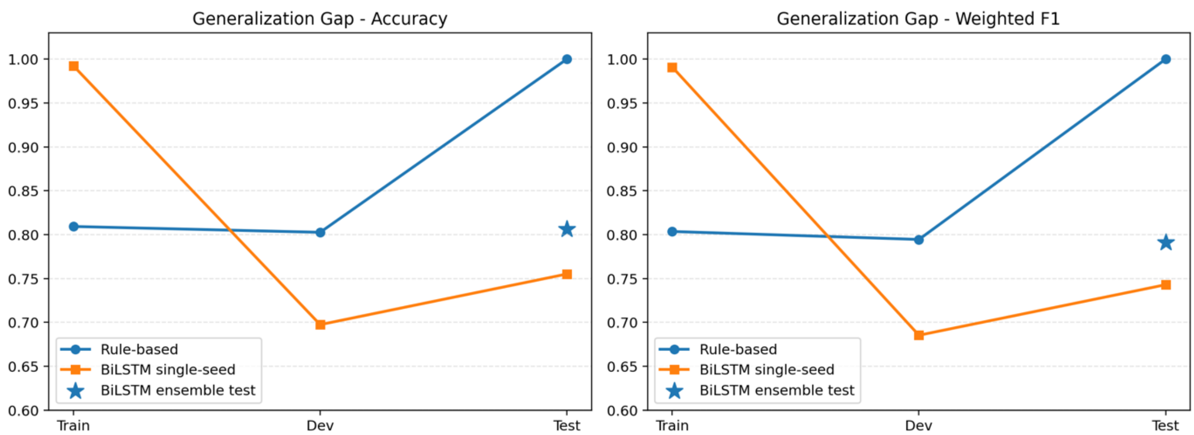

Figure 5. Generation Gap

6.3 Qualitative Examples

Qualitative analysis further illustrates the strengths and limitations of different approaches. In the sentence fragment "…怕你 *miss* 了", the target word "miss" is correctly labeled as a verb in the gold standard. The PyCantonese baseline misclassifies it as a noun due to its default bias, while both BiLSTM and mBERT correctly predict VERB. This demonstrates that neural models are capable of capturing local syntactic cues, such as the aspect marker "了", which signals verbal usage.

In the example "…you *really* made our day!", the word "really" is labeled as an adverb in the gold standard. However, BiLSTM predicts it as a verb, likely influenced by the following verb "made", while mBERT predicts it as an adjective. This case highlights the difficulty of distinguishing between adverbial and adjectival functions in code-mixed informal language, where syntactic and semantic boundaries are less clearly defined.

A more structurally complex example appears in "…一百个 *Like* 赞你哋…", where "Like" functions as a countable noun. The baseline incorrectly labels it as a proper noun, while both neural models classify it as a verb, reflecting their bias toward the more frequent predicate usage of "like" in English. This error reveals a key limitation: neural models fail to incorporate Cantonese-specific syntactic constraints, such as the classifier construction "一百个 + noun", which strongly enforces a nominal interpretation.

Overall, these examples demonstrate that neural models excel at capturing local contextual patterns but lack explicit modeling of Cantonese grammatical structures, while rule-based methods effectively encode such constraints but depend heavily on coverage. This suggests that future improvements should explore hybrid approaches that integrate neural representations with linguistically informed constraints to better handle code-mixed ambiguity.

Table 6. Qualitative comparison of system outputs on selected test examples.

Sentence	Word	Gold	PyCantonese	BiLSTM	mBERT
…怕你 *miss* 了	miss	VERB	NOUN	VERB	VERB
…你们 *really* made our day!	really	ADV	NOUN	VERB	PROPN
…一百個 *Like* 讚你哋…	Like	NOUN	PROPN	VERB	PROPN

7. Conclusion and Future Work

This project investigated POS tagging ambiguity of English words embedded in Cantonese-English mixed text. Four computational approaches were tested: a PyCantonese baseline, a rule-based disambiguation system, a BiLSTM-CRF model, and an mBERT fine-tuning model. The results show that existing Cantonese NLP tools such as PyCantonese perform poorly on English tokens in code-switching contexts, achieving only around 48% accuracy on the test set. Its main weakness is the tendency to default to `NOUN` or `PROPN`, especially when English words are actually used as verbs, adjectives, adverbs, discourse particles, or internet expressions.

Among the four approaches, the rule-based system achieved the best overall performance, reaching 100% accuracy, precision, recall, and F1 on the test set. However, this result should be interpreted with caution, since many of the rules were designed based on observed error patterns in the dataset. Therefore, the perfect score may partly reflect dataset-specific optimisation rather than full generalisability. Nevertheless, the rule-based system was still important because it showed that Cantonese syntactic cues and manually constructed lexicons can effectively resolve many POS ambiguities. These rule-based features also helped improve the later neural models.

The BiLSTM-CRF model improved over the PyCantonese baseline, but its performance was still limited by the small dataset size and by confusion between major categories such as `NOUN`, `PROPN`, and `VERB`. The final mBERT model achieved the strongest performance among the machine-learning approaches. After adding lexicon features and contextual rule features, it reached 90.82% accuracy and 0.853 F1. This shows that multilingual transformer models can be effective for Cantonese-English POS tagging, but plain fine-tuning alone is not enough when the dataset is very small.

Overall, the findings suggest that automatic POS tagging tools alone are insufficient for resolving English POS ambiguity in Cantonese-English mixed text. The best-performing systems all incorporated Cantonese grammatical knowledge and manually constructed lexicons. This supports the central argument of the project: the POS of an English word in code-switching contexts cannot be determined from the English word form alone, but must be interpreted within the surrounding Cantonese syntactic frame.

Several limitations remain. First, the dataset is still small, with only 310 cleaned sentences and 569 evaluable English tokens. This restricts the generalisation ability of neural models, especially mBERT. Second, the gold standard was mainly produced through manual correction by one annotator, so future work should include inter-annotator agreement to improve annotation reliability. Third, although the rule-based system performs extremely well on the current test set, its coverage may decrease when applied to larger, more complex, and more diverse real-world data.

Future work could proceed in three directions. First, the corpus should be expanded to several thousand sentences and include a wider range of online sources. Second, Cantonese-specific pre-trained language models, such as HK-BERT or other Hong Kong Cantonese language models, could be tested. Third, the annotation could be extended from English tokens to all tokens in the sentence, allowing the development of a fully bilingual Cantonese-English POS tagging system. This would make the system more useful for downstream NLP tasks such as machine translation, sentiment analysis, and information retrieval.

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Annexures

All code, datasets, annotated files, and supplementary materials are available in the project GitHub repository: https://github.com/yanlingli031205-wq/CBS5502-code-switching-pos

Annex A — PyCantonese Error Analysis Summary

Category	Error Count	Typical Examples
VERB mistagged as PROPN	18	Delay, share, Update, Quit, compare
INTJ mistagged as PROPN	9	haha, wow, fighting, congrats, Salute
Particles / internet slang mistagged as PROPN	8	lor, btw, XDD, la, pls
VERB mistagged as NOUN	14	follow, love, support, mark, glowing
ADJ / ADV mistagged as NOUN	8	gorgeous, fun, good, really, anyway
Segmentation / multi-word issues	7	part-time, Elon Musk, Now You See Me
Other mistaggings	7	everyone (PRON), over (ADP), yet (ADV)
Total	**71**	

Annex B — Rule-based Tagger: 13-Tier Priority Chain

Priority	Rule	Condition	Predicted Tag
1	INTJ Lexicon	Token in INTJ word list	INTJ
2	X Lexicon	Token in X word list (particles, slang)	X
2b	XD Regex	Matches xd/xdd/xddd… pattern	X
3	PROPN Lexicon	Token in known proper noun list	PROPN
4	ALL-CAPS rule	All-caps; lowercase form not in any lexicon	PROPN
4b	ALL-CAPS override	All-caps; lowercase form in ADJ/VERB/ADV/NOUN	that POS
5	Digit rule	Token is purely numeric	NUM
6	Hyphen rule	Token contains "-"	NOUN
7	Cantonese-context (C1–C6)	Ambiguous token; see Table 2 in main text	context-dependent
8	VERB Lexicon	Token in VERB word list	VERB
9	ADJ Lexicon	Token in ADJ word list	ADJ
10	ADV Lexicon	Token in ADV word list	ADV
11	NOUN Lexicon	Token in NOUN word list	NOUN
12	Initial-capital heuristic	Capitalised; not in any lexicon	PROPN
13	Default	No rule matched	NOUN

Annex C — Lexicon Overview

Lexicon	Approx. Entries	Sample Words
INTJ	~25	haha, wow, wah, fighting, congrats, lol, omg, salute
X	~20	lor, la, wor, btw, pls, hea, cos, XDD, FD
PROPN	~45	apple, google, youtube, ig, BBC, Singapore, Grace

VERB	~60	post, follow, share, keep, miss, delay, support, update
ADJ	~45	cute, chill, dirty, facial, chur, good, clear, fit
ADV	~15	really, anyway, yet, so, besides, non
NOUN	~100+	video, channel, fans, app, booking, feel, link, topic

Annex D — Rule-based Training Set Results (data/train.conll)

216 sentences, 388 tokens — Accuracy: 80.93% (314/388)

POS Tag	Precision	Recall	F1	Support
ADJ	0.8000	0.8511	0.8247	47
ADV	0.5000	0.3333	0.4000	6
CONJ	0.0000	0.0000	0.0000	1
INTJ	0.5333	0.8000	0.6400	10
NOUN	0.7718	0.9127	0.8364	126
NUM	0.0000	0.0000	0.0000	2
PROPN	0.8333	0.7812	0.8065	96
VERB	0.9107	0.7846	0.8430	65
X	0.9583	0.6970	0.8070	33
Macro F1	—	—	**0.5158**	—
Weighted F1	—	—	**0.8036**	—

Note: Macro F1 is low due to CONJ (n=1) and NUM (n=2) receiving zero scores — these rare categories are absent from the lexicons. Weighted F1 (0.80) is the more representative metric as it accounts for class frequency.

Annex E — Work Division

Dominant data collection **Data cleaning and preprocessing** **Name**	**Work Content**
LI Yanling (leader)	Rule-based part; manually annotation with making golden standard
PAN Yiting	Data Collection; Data cleaning; PyCantonese baseline part; Cross model Evaluation
ZENG Xueyi	BiLSTM-CRF model training & adjusting
SHEN Ziqi	mBERT model training & adjusting
ZHANG Qiyi	Dominant data collection & Data cleaning and preprocessing